OPT2026: 18th Annual Workshop on Optimization for Machine Learning

What Is Richer Optimization Geometry Worth? Hessian-Blind Pricing of Structured Preconditioners

Yutian Chu b23090129@njupt.edu.cn
Nanjing University of Posts and Telecommunications

Abstract

Richer preconditioners can reduce the number of optimization steps while costing more to construct and apply. We ask a narrower, candidate-specific question: given a baseline and one frozen richer preconditioner, what compute premium is that candidate locally worth? For a stochastic quadratic model we derive the exact optimal one-step progress $G^\star(P)$, its invariance to scalar rescaling, and the break-even premium $q_\star = G^\star(P_1)/G^\star(P_0)$. We estimate the components of this price without revealing the population Hessian, using independent cross-fitted gradients and stochastic secants. On 1,728 analytically calibrated paired-spectrum instances, predicted prices track constructed-candidate oracle prices (Spearman 0.993; log-price MAE 0.007). Across an offline compute-price sweep, gain-only routing systematically over-purchases, while value routing nearly eliminates decision regret on this benchmark. However, after charging valuation probes, only 6.23% of the 10,368 instance-price rows remain oracle-positive under the modeled costs. This synthetic, one-step diagnostic does not establish multi-step training gains or measured speedups.

1. Introduction

Structured preconditioners trade arithmetic for geometry. Adam-like diagonal methods are cheap to apply, while K-FAC-, Shampoo-, SOAP-, and Muon-style matrix methods exploit richer structure at additional construction or application cost [3, 9, 12, 16, 20]. The relevant comparison is therefore not only whether a richer candidate improves local progress, but whether the improvement justifies its compute premium. We isolate this question for one frozen candidate.

We formalize this question as local pricing. Let P_0 be a baseline and P_1 a richer candidate that has already been constructed and frozen. The largest effective compute multiplier justified by the candidate is

$$q_\star(P_1; P_0) = \frac{G^\star(P_1)}{G^\star(P_0)}, \tag{1}$$

defined when $G^\star(P_0) > 0$. If the construction-amortized price q_{eff} is below $q_\star$, we prefer P_1; otherwise we keep the baseline. The same local gain ratio can therefore justify one modeled cost and fail to justify another.

We derive this price exactly for a stochastic quadratic, establish scale invariance, and characterize its noise dependence for fixed candidates. Our estimator is *Hessian-blind*: construction and pricing use stochastic gradients and secants, with no access to the population Hessian. Controlled synthetic experiments test whether the estimated price tracks the value of the constructed candidate and separates worthwhile purchases from raw progress gains. We report candidate-recovery error and probe overhead separately; the resulting diagnostic concerns local progress, not full training trajectories.

© Y. Chu.

2. Local progress and compute price

Condition on the current filtration. For displacement s, suppose the local objective is exactly

$$f(w+s) = f(w) + h^\top s + \frac{1}{2} s^\top K s, \qquad K \succ 0. \tag{2}$$

Let g satisfy $\mathbb{E}[g] = h$ and $\mathrm{Cov}(g) = \Omega$. A *predictable* SPD preconditioner P is fixed from past information before the current stochastic gradient is drawn, and gives the update $s = -\eta P g$. This includes a frozen positive diagonal map, but not standard Adam's full update: momentum and a denominator that depends on the current gradient need not have this form. The normalized baseline $P_0 = I$ used below is not an implementation of Adam.

Proposition 1 (Optimal local progress) *Define*

$$A_P = h^\top P h, \qquad B_P = h^\top P K P h + \mathrm{tr}(K P \Omega P), \tag{3}$$

and assume $B_P > 0$ (automatic if $h \neq 0$). Then

$$G(\eta, P) := \mathbb{E}[f(w) - f(w+s)] = \eta A_P - \frac{\eta^2}{2} B_P. \tag{4}$$

Consequently,

$$\eta^\star = \frac{A_P}{B_P}, \qquad G^\star(P) = \frac{(h^\top P h)^2}{2\{h^\top P K P h + \mathrm{tr}(K P \Omega P)\}}. \tag{5}$$

Equation (5) gives $G^\star(cP) = G^\star(P)$ for every $c > 0$. The scalar step size therefore absorbs any global rescaling of P; $q_\star$ prices the *shape* of the preconditioning geometry rather than an arbitrary learning-rate scale, echoing norm-based views of optimizer geometry [2]. The price is local: it does not imply a wall-clock guarantee or persistence of the quadratic model along a trajectory.

Let $c_0, c_1 > 0$ be baseline and rich step costs in the same compute units, $C_{\mathrm{build}} \geq 0$ a one-time construction cost, and $T > 0$ the intended number of reuse steps. We use

$$q_{\mathrm{eff}} = \frac{c_1}{c_0} + \frac{C_{\mathrm{build}}/c_0}{T}, \qquad \text{prefer } P_1 \text{ iff } q_\star > q_{\mathrm{eff}}. \tag{6}$$

For $G_j = G^\star(P_j)$, $G_0 > 0$, $R = G_1/G_0$, and $q = c_1/c_0$, the cost-amortization threshold is

$$T_{\mathrm{BE}} = \frac{C_{\mathrm{build}}/c_0}{R - q}, \qquad R > q. \tag{7}$$

Here C_{build}/c_0 is measured in baseline-step units and R, q are dimensionless. Strict preference requires $T > T_{\mathrm{BE}}$; when $R \leq q$, no finite horizon gives strict preference (if $R = q$ and $C_{\mathrm{build}} = 0$, all horizons tie). Equation (7) holds the one-step ratio R fixed while amortizing a cost. It is not a multi-step performance guarantee: gradients, curvature, and noise can change along a trajectory even when P is frozen. Gradient/secant probes used to estimate $q_\star$ are charged separately in the end-to-end diagnostic, so accurate valuation need not have positive deployment value once measurement is included.

2.1. Hessian-blind component estimation

The estimator never receives the population matrix K. It sees independent stochastic gradients and a stochastic secant operator $Y(v)$ satisfying

$$\mathbb{E}[Y(v) \mid v] = Kv. \tag{8}$$

For independent, identically distributed gradients $g_a, \dots, g_f$ with covariance Ω, and fresh secant samples, define

$$\widehat{A}_P = g_a^\top P g_b, \tag{9}$$

$$\widehat{S}_P = (Pg_c)^\top Y(Pg_d), \tag{10}$$

$$\widehat{N}_P = (P\zeta)^\top Y(P\zeta), \qquad \zeta = \frac{g_e - g_f}{\sqrt{2}}. \tag{11}$$

Lemma 2 (Hessian-blind component estimates) *Condition on a frozen P and mutually independent evaluation streams. Then*

$$\mathbb{E}[\widehat{A}_P \mid P] = h^\top P h, \tag{12}$$

$$\mathbb{E}[\widehat{S}_P \mid P] = h^\top P K P h, \tag{13}$$

$$\mathbb{E}[\widehat{N}_P \mid P] = \operatorname{tr}(KP\Omega P). \tag{14}$$

We plug component means into (5); the nonlinear ratio is not claimed unbiased. A delete-one jackknife with a normal critical value gives the empirical lower-confidence rule used below. We treat this as a conservative heuristic: it has no finite-sample coverage guarantee and is not called certified. Discovery, projected-curvature estimation, and value evaluation use disjoint sample streams. The component identities concern update covariance Ω; when evaluation and update batch sizes differ, the noise estimate is rescaled as detailed in Appendix A.1.

Routing rules. Let $q_{\mathrm{LCB}} = \widehat{q}_\star - z_{0.95}\widehat{\mathrm{se}}$, where $z_{0.95} \approx 1.64485$ and $\widehat{\mathrm{se}}$ is the delete-one jackknife standard error. The three rules use the following strict purchase thresholds:

Gain-only:	$\widehat{q}_\star > 1$,
Value:	$\widehat{q}_\star > q_{\mathrm{eff}}$,
LCB Value:	$q_{\mathrm{LCB}} > q_{\mathrm{eff}}$.

Ties and invalid estimates select the baseline; a nonfinite lower bound also prevents LCB purchase. Gain-only ignores cost, Value compares estimated local gain per cost, and LCB Value adds an uncertified uncertainty margin.

For the concrete low-rank candidate used in the experiments, we work in baseline-normalized coordinates with $P_0 = I$. Let W be a probe matrix and let an independent discovery batch return a stochastic secant matrix $Y_b(W)$ with $\mathbb{E}[Y_b(W) \mid W] = KW$. Residual secants target the geometry missed by the baseline, following the broader randomized-sketching view of low-rank curvature approximation [8, 10, 17]:

$$Q = \operatorname{orth}(Y_b(W) - W). \tag{15}$$

A separate stream observes $Y_c(Q)$. Let $\widetilde{K}_S$ be $\operatorname{sym}(Q^\top Y_c(Q))$ after adding a fixed positive ridge and clipping its eigenvalues to a positive interval. Then

$$P_1 = I + Q(\widetilde{K}_S^{-1} - I)Q^\top. \tag{16}$$

Positive clipping makes P_1 SPD. The candidate is frozen before value evaluation; the planted ideal inverse is a post hoc diagnostic reference.

2.2. Noise is candidate-specific

For fixed $h \neq 0$, $K, P_0, P_1 \succ 0$, and $\Omega(n) = n\Omega_0$, $n \geq 0$, with $\Omega_0 \succeq 0$, write $A_j = h^\top P_j h$ and

$$S_j = h^\top P_j K P_j h, \qquad N_j = \operatorname{tr}(K P_j \Omega_0 P_j), \qquad \chi_j = N_j / S_j. \tag{17}$$

Proposition 3 (Candidate-specific noise effect) *The break-even premium satisfies*

$$q_\star(n) = \frac{A_1^2}{A_0^2} \frac{S_0 + nN_0}{S_1 + nN_1}, \qquad \operatorname{sign} q_\star'(n) = \operatorname{sign}(\chi_0 - \chi_1). \tag{18}$$

Hence the effect of noise on the relative value of the richer candidate is candidate-dependent.

3. Analytically calibrated benchmark

Synthetic v1 shows that a broad random family can be dominated by cases in which the richer candidate is not worth its price. To evaluate the pricing boundary itself rather than class imbalance, Synthetic v2 uses an analytically calibrated family

$$K = I + U \Lambda U^\top, \tag{19}$$

with even rank r, random orthonormal U, and $\Lambda = \operatorname{diag}(+a, -a, \ldots)$, $0 < a < 1$. The unit gradient places fraction γ of squared energy symmetrically in the shifted modes; a diffuse orthogonal residual of strength ϵ controls mismatch. For zero noise/mismatch and exact recovery,

$$h^\top K h = 1, \qquad h^\top K^{-1} h = 1 + \frac{\gamma a^2}{1 - a^2}, \tag{20}$$

so

$$q_\star^{\text{ideal}} = 1 + \frac{\gamma a^2}{1 - a^2}. \tag{21}$$

Tests reproduce (21) to 2.3×10^{-14}. This construction creates controlled instances on both sides of the price boundary without conditioning on observed outcomes.

We fix $d = 64$; $r \in \{4, 8\}$; $a \in \{.2, .5, .8, .9\}$; $\gamma \in \{.15, .40, .65, .90\}$; mismatch $\epsilon \in \{0, .10, .25\}$; noise scale $n \in \{0, .20, 1\}$; and six deterministic seeds, for 1,728 base instances. The formal run uses 4,096 discovery and curvature samples. We also report a 128/256-sample low-probe ablation. The 144 geometry-specific candidates are reused across gradient-energy and noise settings; each resulting base state is evaluated once. Prices $\{1.05, 1.25, 1.5, 2, 3, 4\}$ are swept offline without changing candidates or evaluation samples.

4. Results

Across base instances, $\widehat{q}_\star$ has Spearman correlation 0.9926 with oracle $q_\star$, log-price MAE 0.0070, and zero invalid estimates (Figure 1a). This is an instance-level continuous pricing result rather than a classification accuracy measure. Oracle-positive fractions vary naturally with cost: 0.490, 0.189,

Table 1: Routing over 1,728 base instances per price. Precision and recall concern richer-candidate purchases; regret is oracle local decision regret.

	Value		LCB Value		Mean regret	
q_{eff}	Prec.	Rec.	Prec.	Rec.	Gain-only	Value
1.05	.972	.985	.997	.909	.00686	.00005
1.25	.985	.988	1.000	.877	.05267	.00002
1.50	.979	.959	1.000	.871	.10608	.00003
2.00	1.000	.838	1.000	.676	.18242	.00003

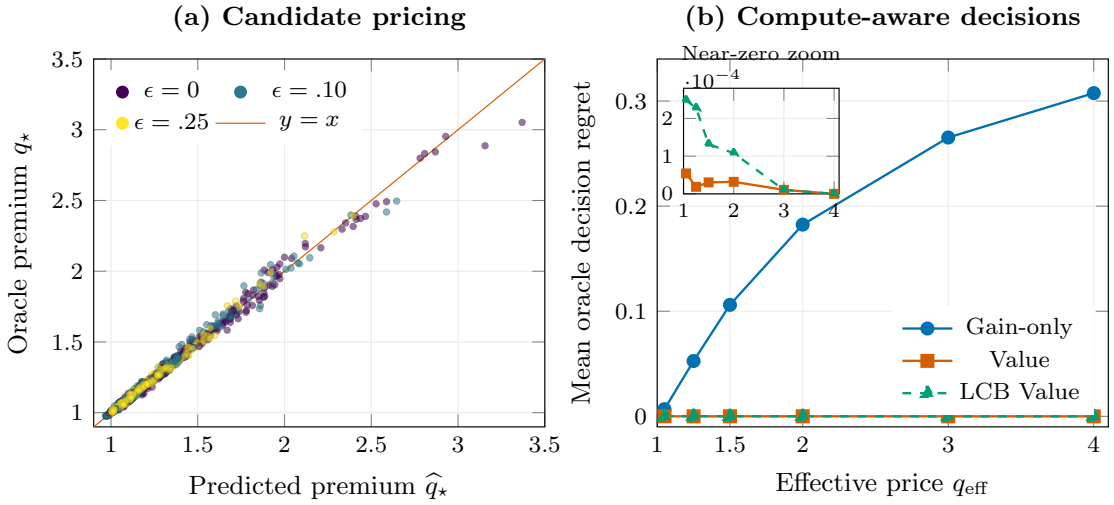


Figure 1: Synthetic v2. (a) Predicted versus oracle break-even premium; color denotes mismatch ϵ. (b) Oracle decision regret versus compute price. The inset expands the near-zero Value and LCB Value curves; the main linear scale retains Gain-only's full regret.

0.085, and 0.021 at $q_{\text{eff}} = 1.05$, 1.25, 1.5, and 2.0. The oracle always evaluates the *constructed* candidate, not the planted ideal inverse.

Gain-only routing selects the richer candidate on 90.9% of base instances because it asks only whether estimated progress improves. Across the six offline price levels, 8,165 instance-price rows satisfy

$$1 < q_\star < q_{\text{eff}}, \tag{22}$$

so the richer candidate improves raw local progress but is still too expensive. Gain-only purchases the richer candidate in 7,919 of these rows.

Table 1 compares the routing rules. At $q_{\text{eff}} = 2$, Value reduces mean regret from 0.18242 to 0.000032; LCB regret is 0.000108, with zero false purchases and 67.6% recall. Regret compares $\max(G_0, G_1/q_{\text{eff}})$ with the policy's selected value; it measures local decision loss, not training loss over a trajectory. Price rows and states sharing a candidate are dependent; we make no independent-sample inference from them.

Valuation overhead. Charging the gradient/secant evaluation probes adds 0.1792 baseline-step-normalized cost per reuse step, giving $q_{\mathrm{e2e}} = q_{\mathrm{eff}} + 0.1792$ under the stated 10,000-step amortization model. Only $646/10{,}368 = 6.23\%$ of instance-price rows satisfy $q_\star > q_{\mathrm{e2e}}$. This is the oracle worthwhile-purchase fraction under modeled costs, not the success rate of an implemented optimizer or evidence of real training speedup. Accurate local pricing can therefore coexist with an expensive valuation procedure. Appendix A.1 gives the sample charges; the sweep fixes total q_{eff}, not measured per-step runtime ratios.

Mismatch increases mean construction gap from 0.313 at $\epsilon = 0$ to 0.371 at 0.25, while mean subspace error is 0.582 (Figure 2). The 128/256-sample ablation is harder (0.857 subspace error) and produces almost no positive cases beyond price 1.05. Equation (18) predicts every valid frozen-candidate noise-slope sign.

Synthetic v1 provides a complementary negative stress test. Only $12/1{,}728$ price rows are oracle-positive, so its 0.9936 sign accuracy is uninformative: Always Baseline already obtains 0.9931. More informative are the continuous ranking (Spearman 0.966), Value regret 0.0046 versus 0.0060 for Always Baseline, and 526 gain-only purchases with $1 < G_1/G_0 < c_1/c_0$ that Value rejects. Thus arbitrary richer geometry often fails to justify its modeled compute cost.

5. Related work and limitations

Diagonal baselines are inexpensive [4, 6, 12, 15]; richer curvature methods [9, 13, 16, 22] use factorization, scaling, or refresh to control cost [1, 7, 19, 20]. Muon need not meet our frozen-SPD assumptions [3, 14], and unified analysis covers broader structures [21]. Other selection methods use scout trajectories [18], sketched diagnostics [5], or online trajectory optimization [11]. We instead price one frozen candidate locally; the named optimizers motivate the question but are not implemented baselines in our experiments.

The evidence is synthetic-only and the theory is exact for the stated quadratic model. We provide no nonquadratic remainder bound, multi-step reuse guarantee, or measured training-speed comparison. The cost model exposes valuation overhead but does not measure hardware runtime, and the jackknife LCB has no finite-sample or sequential coverage guarantee. Testing an implemented pricing policy on nonquadratic training tasks with measured construction, application, and probe costs remains necessary before deployment claims.

References

[1] Rohan Anil, Vineet Gupta, Tomer Koren, Kevin Regan, and Yoram Singer. Scalable second order optimization for deep learning. *arXiv preprint arXiv:2002.09018*, 2020. URL https://arxiv.org/abs/2002.09018.

[2] Jeremy Bernstein and Laker Newhouse. Old optimizer, new norm: An anthology. In *NeurIPS 2024 Workshops: OPT*, 2024. URL https://www.opt-ml.org/papers/2024/paper93.pdf.

[3] Lizhang Chen, Jonathan Li, and Qiang Liu. Muon optimizes under spectral norm constraints. In *NeurIPS 2025 Workshops: OPT*, 2025. URL https://www.opt-ml.org/papers/2025/paper137.pdf.

[4] Alexandre Défossez, Leon Bottou, Francis Bach, and Nicolas Usunier. A simple convergence proof of adam and adagrad. *Transactions on Machine Learning Research*, 2022. URL https://openreview.net/forum?id=aIoEkwc2oB.

[5] Conner DiPaolo and Weiqing Gu. A randomized algorithm for preconditioner selection. *arXiv preprint arXiv:1908.00633*, 2019. URL https://arxiv.org/abs/1908.00633.

[6] John Duchi, Elad Hazan, and Yoram Singer. Adaptive subgradient methods for online learning and stochastic optimization. *Journal of Machine Learning Research*, 12:2121–2159, 2011.

[7] Runa Eschenhagen, Aaron Defazio, Tsung-Hsien Lee, Richard E. Turner, and Hao-Jun Shi. Purifying shampoo: Investigating shampoo's heuristics by decomposing its preconditioner. In *Advances in Neural Information Processing Systems*, volume 38, 2025. doi: 10.52202/085713-5507. URL https://papers.nips.cc/paper_files/paper/2025/hash/f14f4eda29a74c02c803699a09529bb9-Abstract-Conference.html.

[8] Zachary Frangella, Pratik Rathore, Shipu Zhao, and Madeleine Udell. SketchySGD: Reliable stochastic optimization via randomized curvature estimates. *SIAM Journal on Mathematics of Data Science*, 6(4): 1173–1204, 2024. doi: 10.1137/23M1575330.

[9] Vineet Gupta, Tomer Koren, and Yoram Singer. Shampoo: Preconditioned stochastic tensor optimization. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 1842–1850. PMLR, 2018. URL https://proceedings.mlr.press/v80/gupta18a.html.

[10] Nathan Halko, Per-Gunnar Martinsson, and Joel A. Tropp. Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Review*, 53(2):217–288, 2011. doi: 10.1137/090771806.

[11] Haichen Hu and David Simchi-Levi. Optimizing the preconditioner: A black-box online-to-nonconvex conversion with static regret minimization oracles. *arXiv preprint arXiv:2607.17607*, 2026. URL https://arxiv.org/abs/2607.17607v2.

[12] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015. URL https://arxiv.org/abs/1412.6980.

[13] Hong Liu, Zhiyuan Li, David Hall, Percy Liang, and Tengyu Ma. Sophia: A scalable stochastic second-order optimizer for language model pre-training. In *International Conference on Learning Representations*, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/06960915ba8674c7a898ec0b472b80ff-Abstract-Conference.html.

[14] Jingyuan Liu, Jianlin Su, Xingcheng Yao, Zhejun Jiang, Guokun Lai, Yulun Du, Yidao Qin, Weixin Xu, Enzhe Lu, Junjie Yan, Yanru Chen, Huabin Zheng, Yibo Liu, Shaowei Liu, Bohong Yin, Weiran He, Han Zhu, Yuzhi Wang, Jianzhou Wang, Mengnan Dong, Zheng Zhang, Yongsheng Kang, Hao Zhang, Xinran Xu, Yutao Zhang, Yuxin Wu, Xinyu Zhou, and Zhilin Yang. Muon is scalable for llm training. *arXiv preprint arXiv:2502.16982*, 2025. URL https://arxiv.org/abs/2502.16982.

[15] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019. URL https://openreview.net/forum?id=Bkg6RiCqY7.

[16] James Martens and Roger Grosse. Optimizing neural networks with kronecker-factored approximate curvature. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 2408–2417. PMLR, 2015. URL https://proceedings.mlr.press/v37/martens15.html.

[17] Mert Pilanci and Martin J. Wainwright. Newton sketch: A near linear-time optimization algorithm with linear-quadratic convergence. *SIAM Journal on Optimization*, 27(1):205–245, 2017. doi: 10.1137/15M1021106.

[18] Ronald Richman and Mario V. Wüthrich. Many optimizers but only one training path: Repeated resampling for adaptive optimizer selection. *arXiv preprint arXiv:2608.18810*, 2026. URL https://arxiv.org/abs/2608.18810.

[19] Noam Shazeer and Mitchell Stern. Adafactor: Adaptive learning rates with sublinear memory cost. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 4596–4604. PMLR, 2018. URL https://proceedings.mlr.press/v80/shazeer18a.html.

[20] Nikhil Vyas, Depen Morwani, Rosie Zhao, Itai Shapira, David Brandfonbrener, Lucas Janson, and Sham M. Kakade. SOAP: Improving and stabilizing shampoo using adam for language modeling. In *International Conference on Learning Representations*, 2025. URL https://openreview.net/forum?id=IDxZhXrpNf.

[21] Shuo Xie, Tianhao Wang, Sashank J. Reddi, Sanjiv Kumar, and Zhiyuan Li. Structured preconditioners in adaptive optimization: A unified analysis. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 68746–68776. PMLR, 2025. URL https://proceedings.mlr.press/v267/xie25j.html.

[22] Zhewei Yao, Amir Gholami, Sheng Shen, Mustafa Mustafa, Kurt Keutzer, and Michael W. Mahoney. ADAHESSIAN: An adaptive second order optimizer for machine learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 10665–10673, 2021. doi: 10.1609/aaai.v35i12.17275.

Appendix A. Additional diagnostics and reproducibility

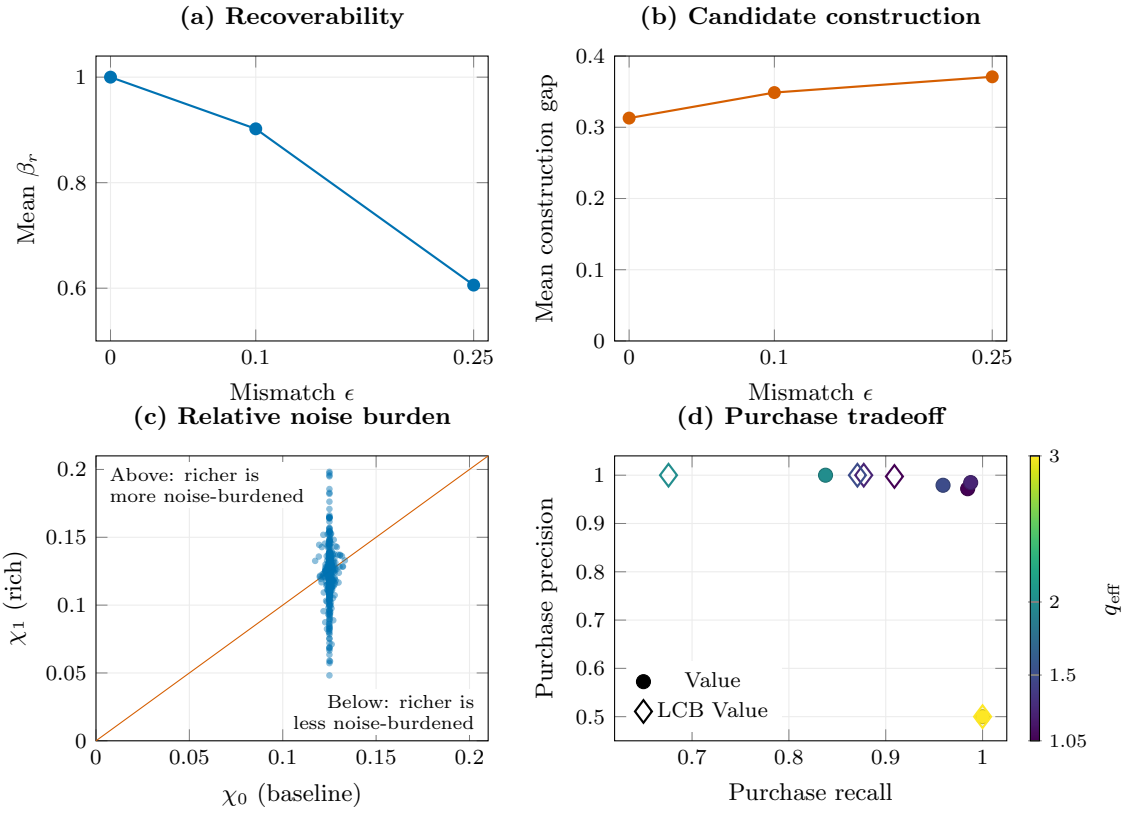


Figure 2: Mechanism diagnostics from frozen raw results. (a) Recoverable residual energy. (b) Gap to the planted reference. (c) Above $y = x$, noise lowers the fixed candidate's relative price; below, it raises the price. (d) Marker shape denotes policy and color denotes effective price. At price 3 both policies coincide; price 4 has no purchases or positives, so precision and recall are undefined and omitted. No points are jittered.

For each base instance, discovery, projected-curvature estimation, and value evaluation use distinct deterministic seeds. The estimator interface contains no population Hessian, planted basis, spectral amplitude, gradient-energy fraction, or recoverability field. Price predictions and uncertainty estimates are frozen before a phase-gated oracle reveals population quantities; routing then applies fixed thresholds offline. Tests cover this gate, stream determinism, $G^\star(cP) = G^\star(P)$, the paired-spectrum identity, cost algebra, monotone price sweeps, and output schema. The formal v2 run produced 1,728 base rows and 10,368 offline price rows. Its results, the low-probe ablation, and the separate v1 stress test are traced to distinct frozen files.

A.1. Evaluation protocol

The constructor and estimator receive stochastic sampling methods, not population K or planted U. For v2, the sampling service returns gradients with mean h and individual covariance nI/d. For a matrix V, it returns $Y_b(V) = X^\top XV/b$, where the independent rows of X have distribution $\mathcal{N}(0, K)$. Consequently $\mathbb{E}[Y_b(V) \mid V] = KV$. Population quantities inside this synthetic sampling service are not visible through the estimation interface.

Algorithm 1: Frozen v2 construction and pricing protocol. The low-probe ablation replaces the two construction batch sizes by 128 and 256.

Input: Sampling interface, fixed rank r, independent discovery/curvature/evaluation streams; $P_0 = I$.
Draw $W \in \{-1/\sqrt{r}, +1/\sqrt{r}\}^{d\times r}$ using the discovery stream
$Q \leftarrow \mathrm{qr}(Y_{4096}(W) - W)$ using that stream
$M \leftarrow \mathrm{sym}(Q^\top Y_{4096}(Q))$ using the curvature stream
Diagonalize $M = V\,\mathrm{diag}(\lambda)V^\top$; set $\widetilde{\lambda}_i = \mathrm{clip}(\lambda_i + .005, .05, 2)$
$\widetilde{K}_S \leftarrow V\,\mathrm{diag}(\widetilde{\lambda})V^\top$; freeze $P_1 \leftarrow I + Q(\widetilde{K}_S^{-1} - I)Q^\top$
For 64 repetitions, use fresh evaluation batches to estimate (A_j, S_j, N_j) for both candidates, with the covariance correction below
Form component means, $\widehat{q}_\star$, and its delete-one jackknife lower bound; freeze these outputs
Seal predictions, then reveal the oracle once for diagnostics
Apply fixed price thresholds offline to the saved estimates

Table 2: Statistical roles within one base state. Paired estimates of P_0 and P_1 share evaluation draws; distinct roles use separate streams.

Role	Observed inputs	Frozen output / downstream use
Discovery	W, discovery secants	Q for projected curvature
Curvature	Q, fresh secants	SPD P_1 for valuation
Evaluation	P_0, P_1, fresh gradients/secants	Price and uncertainty for routing
Oracle	Population data after sealing	Labels and diagnostics only

Batches, uncertainty, and guards. Each evaluation repetition draws six independent gradient batches of size $m_e = 16$. Signal and noise each use a fresh secant batch of size 32, applied jointly to the two candidate vectors. The actual update batch size is $m_u = 8$. Writing individual gradient covariance as Σ, evaluation differences have covariance Σ/m_e, whereas the update covariance in Proposition 1 is $\Omega = \Sigma/m_u$. The implemented noise estimate is therefore

$$\widehat{N}_P^{\mathrm{update}} = \frac{m_e}{m_u}(P\zeta)^\top Y(P\zeta) = 2(P\zeta)^\top Y(P\zeta).$$

No rescaling is needed for the signal cross-products. For $m = 64$ delete-one price estimates $\widehat{q}_{(-i)}$, the standard error is

$$\widehat{\mathrm{se}}^2 = \frac{m-1}{m}\sum_i (\widehat{q}_{(-i)} - \overline{q}_{(-\cdot)})^2, \qquad q_{\mathrm{LCB}} = \widehat{q}_\star - 1.6448536269514722\,\widehat{\mathrm{se}}.$$

Value purchases when $\widehat{q}_\star > q_{\mathrm{eff}}$; LCB Value uses $q_{\mathrm{LCB}} > q_{\mathrm{eff}}$; Gain-only uses $\widehat{q}_\star > 1$. The gain helper returns zero for nonpositive estimated A or estimated $B \le 10^{-12}$, and NaN for nonfinite

inputs. Pricing requires finite positive component means and a finite price; invalid estimates or nonfinite lower bounds do not trigger purchase. There were zero invalid prices in the frozen formal run. The jackknife bound is a normal-approximation heuristic, not a coverage certificate.

Grid and reuse. The grid in the main text contains $2 \cdot 4 \cdot 3 \cdot 6 = 144$ geometry candidates indexed by $(r, a, \epsilon, \text{seed})$. Each is reused for four gradient-energy fractions and three noise levels, giving 1,728 base states and 10,368 price decisions. No price is used to fit a candidate. The low-probe run uses the identical grid, evaluation settings, ridge, clipping bounds, and cost model; only discovery/curvature batch sizes change. The results describe both frozen configurations without claiming that the low-probe study was preregistered.

Seed policy. The base seed is 20260906 and replicate indices are $0, \ldots, 5$. Define $k(t) = \text{round}(10000t)$ and let $s(\cdot)$ be the first uint32 output of NumPy SeedSequence given the listed integers. The seed inputs, with $b = 20260906$ and replicate i, are

$$\begin{aligned} \text{problem:} \quad & (b, d, r, k(a), i, 0), \\ \text{discovery/curvature:} \quad & (b, d, r, k(a), k(\epsilon), i, j), \quad j = 1, 2, \\ \text{evaluation:} \quad & (b, d, r, k(a), k(\epsilon), k(\gamma), i, 3). \end{aligned}$$

Distinct role seeds are checked and saved. Problem seeds omit mismatch and evaluation seeds omit noise, intentionally coupling grid states. At zero noise, deterministic gradient calls do not consume random draws, so equal evaluation seeds need not give identical secant samples across zero/nonzero noise. These repeated states are not independent scientific replications.

Costs. A secant call is charged its batch size times the number of input vectors; each gradient call is charged its batch size. One sample costs 0.125 baseline-step units. Formal construction uses $8192r$ samples; evaluation uses $64(6 \cdot 16 + 4 \cdot 32) = 14336$. Both reuse horizons are 10,000. The grid fixes total q_{eff}, with implied $q_{\text{step}} = q_{\text{eff}} - C_{\text{build}}/(10000c_0)$; it does not fix measured per-step ratios. Some implied step ratios are below one even though all total prices exceed one. For the separate end-to-end diagnostic we add evaluation cost, giving $q_{\text{e2e}} = q_{\text{eff}} + .1792$. Exactly $646/10368 = 6.23\%$ of price rows have $q_\star > q_{\text{e2e}}$. This is an oracle worthwhile-purchase fraction under the cost model, not the success rate of an implemented end-to-end training algorithm.

Result definitions and traceability. In v2, let $\mathcal{R} = \{1 < q_\star < q_{\text{eff}}\}$. The 7,919 count is $|\mathcal{R} \cap \{\widehat{q}_\star > 1\}|$: Gain-only purchases within that region. The entire region has 8,165 rows; it is not the reported purchase count. The v1 count 526 additionally requires that Value rejects: Gain-only is true, Value is false, and $1 < G_1/G_0 < c_1/c_0$. It is a particular diagnostic subset, not the total false-purchase count of either policy.

The recovery-run directory contains three formal-v2 tables:

- base states: `synthetic_v2_base_results.csv`;
- price decisions: `synthetic_v2_price_results.csv`;
- metrics: `router_metrics_v2.csv`.

The low-probe run has a distinct directory. The v1 tables are `synthetic_results.csv` and `router_metrics.csv`. Companion traceability records map claims, figure inputs, and configuration hashes to these frozen artifacts. The figures use only saved rows; no new candidate

construction or experimental sampling is involved. The original manifest records Python 3.13.14 and the run command but does not pin a code commit or full dependency lock.

Diagnostic definitions. For singular values s_i of $K - I$ in descending order, recoverability is $\beta_r = \sum_{i \le r} s_i^2 / \sum_i s_i^2$. Subspace error is $1 - \|Q^\top U\|_F^2 / r$. Construction gap is $G^\star(P_{\mathrm{planted}}) - G^\star(P_1)$, where P_{planted} inverts the planted shifted modes and is identity on their complement. This reference is not asserted to maximize noisy progress in every problem. All formal-run gaps happen to be positive; their mean is unchanged. In Figure 2(c), the noise coefficient corresponds to update covariance $nI/(8d)$, so $\chi_0 = 1/(8h^\top Kh)$; the narrow horizontal spread is a property of the benchmark, not missing data.

Appendix B. Complete routing metrics

Every row uses the same 1,728 formal-run base states. With strict oracle label $q_\star > q_{\mathrm{eff}}$ and a policy's purchase decision, precision is $\mathrm{TP}/(\mathrm{TP} + \mathrm{FP})$, recall is $\mathrm{TP}/(\mathrm{TP} + \mathrm{FN})$, false-purchase fraction (FPF) is $\mathrm{FP}/(\mathrm{TP} + \mathrm{FP})$, false-positive rate (FPR) is $\mathrm{FP}/(\mathrm{FP} + \mathrm{TN})$, and missed-opportunity rate (Miss) is $\mathrm{FN}/(\mathrm{TP} + \mathrm{FN})$. Zero-denominator ratios are shown as "–", not zero. Regret is the mean of $\max(G_0, G_1/q_{\mathrm{eff}})$ minus the selected value, in baseline-step-normalized units.

At $q_{\mathrm{eff}} = 3$ only one base state is oracle-positive; both Value policies have one true and one false purchase. At price 4 there are no oracle-positive states and neither Value policy purchases. These sparse high-price cases cannot establish stable precision or recall performance.

Table 3: All prices and policies from the frozen formal v2 evidence. Display rounding only; the saved metrics retain full precision. Baseline and Rich always select their respective candidates.

q_{eff}	Policy	Prec.	Rec.	FPF	FPR	Miss	Regret
1.05	Baseline	–	0.000	–	0.000	1.000	0.055583
1.05	Rich	0.490	1.000	0.510	1.000	0.000	0.009034
1.05	Gain-only	0.539	1.000	0.461	0.821	0.000	0.006860
1.05	Value	0.972	0.985	0.028	0.027	0.015	0.000054
1.05	LCB Value	0.997	0.909	0.003	0.002	0.091	0.000248
1.25	Baseline	–	0.000	–	0.000	1.000	0.024103
1.25	Rich	0.189	1.000	0.811	1.000	0.000	0.061453
1.25	Gain-only	0.208	1.000	0.792	0.887	0.000	0.052673
1.25	Value	0.985	0.988	0.015	0.004	0.012	0.000019
1.25	LCB Value	1.000	0.877	0.000	0.000	0.123	0.000228
1.50	Baseline	–	0.000	–	0.000	1.000	0.009877
1.50	Rich	0.085	1.000	0.915	1.000	0.000	0.120639
1.50	Gain-only	0.094	1.000	0.906	0.900	0.000	0.106078
1.50	Value	0.979	0.959	0.021	0.002	0.041	0.000030
1.50	LCB Value	1.000	0.871	0.000	0.000	0.129	0.000132
2.00	Baseline	–	0.000	–	0.000	1.000	0.001682
2.00	Rich	0.021	1.000	0.979	1.000	0.000	0.204208
2.00	Gain-only	0.024	1.000	0.976	0.907	0.000	0.182422
2.00	Value	1.000	0.838	0.000	0.000	0.162	0.000032
2.00	LCB Value	1.000	0.676	0.000	0.000	0.324	0.000108
3.00	Baseline	–	0.000	–	0.000	1.000	0.000004
3.00	Rich	0.001	1.000	0.999	1.000	0.000	0.294295
3.00	Gain-only	0.001	1.000	0.999	0.909	0.000	0.265284
3.00	Value	0.500	1.000	0.500	0.001	0.000	0.000011
3.00	LCB Value	0.500	1.000	0.500	0.001	0.000	0.000011
4.00	Baseline	–	–	–	0.000	–	0.000000
4.00	Rich	0.000	–	1.000	1.000	–	0.340173
4.00	Gain-only	0.000	–	1.000	0.909	–	0.307549
4.00	Value	–	–	–	0.000	–	0.000000
4.00	LCB Value	–	–	–	0.000	–	0.000000

Appendix C. Proof details

Domains and degenerate cases. All expectations condition on the pre-update information: h, K and the frozen P are fixed, $h \in \mathbb{R}^d$, $K, P \in \mathbb{R}^{d\times d}$ are SPD, and $\Omega \succeq 0$ is finite. The secant randomness is fresh conditional on its input and independent of the left gradient in the signal cross-product. Since

$$\operatorname{tr}(KP\Omega P) = \|K^{1/2}P\Omega^{1/2}\|_F^2,$$

$B_P > 0$ unless both $h = 0$ and $\Omega = 0$. If $h = 0$ but $\Omega \neq 0$, optimal progress is zero at $\eta = 0$ and the price ratio is undefined. Throughout price and noise comparisons we therefore require $h \neq 0$, ensuring $A_j, S_j, G_j > 0$. Costs satisfy $c_0, c_1, T > 0$ and $C_{\text{build}} \geq 0$; equality at a routing threshold selects the baseline.

Proposition 1. Substituting $s = -\eta Pg$ gives

$$\mathbb{E}[f(w) - f(w+s)] = \eta h^\top Ph - \frac{\eta^2}{2}\mathbb{E}[g^\top PKPg]. \tag{23}$$

Since $\mathbb{E}[gg^\top] = hh^\top + \Omega$,

$$\mathbb{E}[g^\top PKPg] = h^\top PKPh + \operatorname{tr}(KP\Omega P), \tag{24}$$

which gives the stated concave quadratic in η and (5). Scale invariance follows because $A_{cP} = cA_P$ and $B_{cP} = c^2 B_P$.

Lemma 2. Conditional on frozen P, independence gives

$$\mathbb{E}[g_a^\top Pg_b \mid P] = \mathbb{E}[g_a]^\top P\mathbb{E}[g_b] = h^\top Ph. \tag{25}$$

Using a fresh secant independent of g_c, g_d,

$$\mathbb{E}[(Pg_c)^\top Y(Pg_d) \mid P] = \mathbb{E}[(Pg_c)^\top K(Pg_d)] \tag{26}$$

$$= h^\top PKPh. \tag{27}$$

Finally, $\zeta = (g_e - g_f)/\sqrt{2}$ has mean zero and covariance Ω under independent identically distributed gradient streams, so

$$\mathbb{E}[(P\zeta)^\top Y(P\zeta) \mid P] = \mathbb{E}[\zeta^\top PKP\zeta] \tag{28}$$

$$= \operatorname{tr}(PKP\Omega) \tag{29}$$

$$= \operatorname{tr}(KP\Omega P). \tag{30}$$

Proposition 3 (candidate-specific noise effect). For fixed h, K, P_0, P_1 and $\Omega(n) = n\Omega_0$,

$$q_\star(n) = \frac{A_1^2}{A_0^2}\frac{S_0 + nN_0}{S_1 + nN_1}. \tag{31}$$

Differentiating gives

$$q_\star'(n) = \frac{A_1^2}{A_0^2}\frac{N_0S_1 - N_1S_0}{(S_1 + nN_1)^2}.$$

The denominator and A_1^2/A_0^2 are positive, while the numerator is $S_0S_1(\chi_0 - \chi_1)$, proving the sign identity, including equality.

Paired-spectrum identity. In the ideal benchmark, the shifted eigenvalues occur in $1 + a, 1 - a$ pairs and the gradient places equal energy in the positive and negative member of each pair. Hence the contribution of the shifted subspace to $h^\top Kh$ averages to one, while its contribution to $h^\top K^{-1}h$ is

$$\frac{1}{2}\left(\frac{1}{1+a} + \frac{1}{1-a}\right) = \frac{1}{1-a^2}. \tag{32}$$

With fraction γ of gradient energy in that subspace and $1 - \gamma$ in unit-curvature directions, we obtain (21).

Break-even horizon. The condition $q_\star > q_{\text{eff}}$ is $R > q + (C_{\text{build}}/c_0)/T$. If $R > q$, rearranging gives $T > T_{\text{BE}}$ with the value stated above; if $R \le q$, no finite reuse horizon gives strict preference. When construction cost is zero and $R = q$, all horizons tie. This rearranges a fixed local-value comparison; it does not sum gains along an evolving optimization trajectory.

Candidate definiteness. For $x = Qu + v$ with $Q^\top v = 0$, we have $x^\top P_1 x = u^\top \widetilde{K}_S^{-1} u + \|v\|^2$. This is positive for nonzero x because the stabilized eigenvalues lie in $[.05, 2]$. Thus P_1 has eigenvalues in $[.5, 20]$ on the discovered subspace and one on its orthogonal complement.